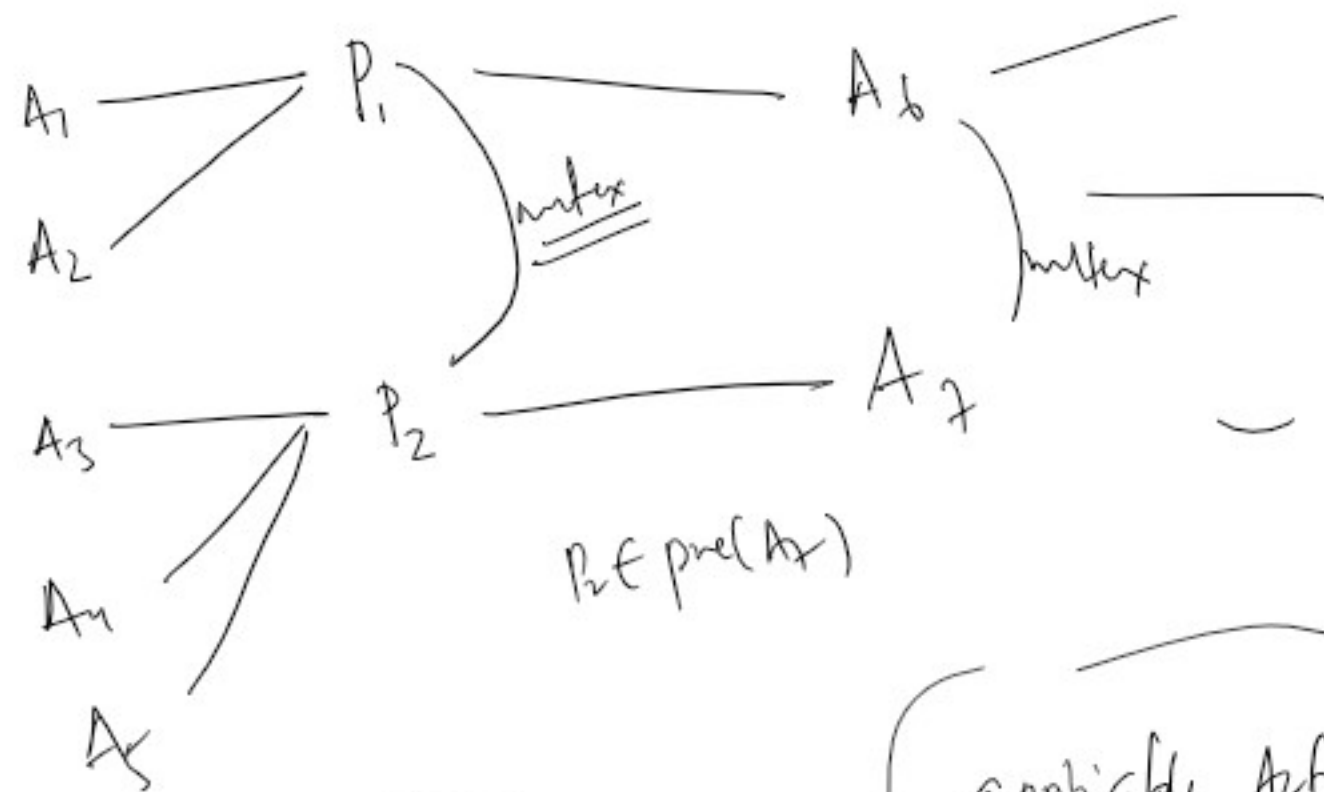
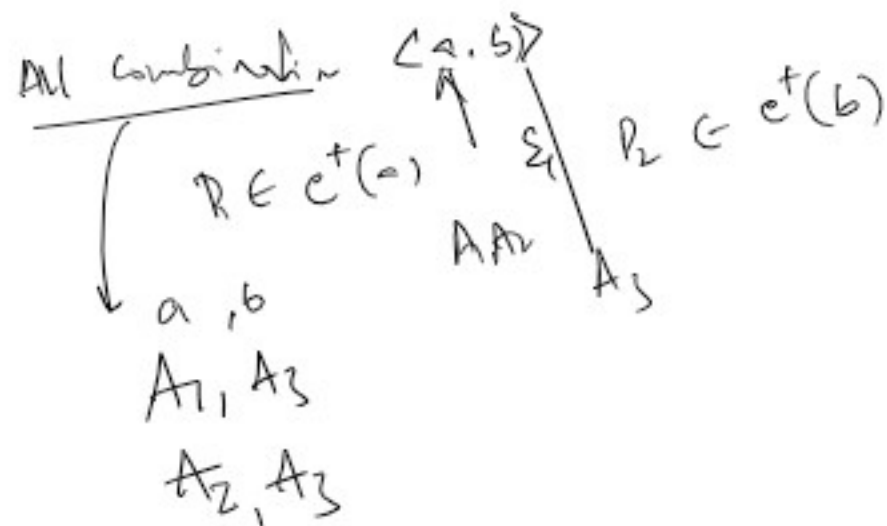


$P_1 \in \text{pre}(A_6)$



$P_2 \in \text{pre}(A_7)$



applicable Action

$\text{pre}(\text{Action}) \in \text{Stack}$

AND

All pairs of preconditions are mutex

Now nil



the
all mutex

Pickup(X) - Pick up X from the table.

Preconditions: { armEmpty, clear(X), onTable(X) }

Add-Effects : { holding(X) }

Del-Effects : { armEmpty, onTable(X) }

Putdown(X) - Place X on the table.

Preconditions: { holding(X) }

Add-Effects : { armEmpty, onTable(X) }

Del-Effects : { holding(X) }

Unstack(X,Y) - Pick up X which is directly on Y.

Preconditions: { armEmpty, clear(X), on(X,Y) }

Add-Effects : { clear(Y), holding(X) }

Del-Effects : { armempty, on(X,Y) }

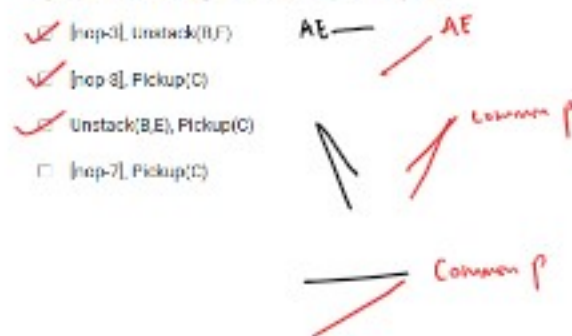
Stack(X,Y) - Place X on Y.

Preconditions: { holding(X), clear(Y) }

Add-Effects : { armEmpty, on(X,Y) }

Del-Effects : { holding(X), clear(Y) }

1) Which of the following are mutex action pairs in Layer 1?



2) Which of the following are mutex proposition pairs in Layer 1?

- ☒ clear(E), holding(C)
☒ clear(B), holding(C)
☒ clear(D), holding(C)
☒ holding(B), holding(C)

3) Which of the following are applicable actions in Layer 2?

- ☒ Pickup(A)
☒ Pickup(C)
☒ Putdown(C)
☒ Stack(B,E)
☐ Stack(C,E)
- $AE, cl(A), m^T(A)$
 $AE, cl(C), m^T(C)$
 $hold(C)$
 $hold(B), clear(E)$
 $\langle un(B,E), un(B,A) \rangle$
 $hold(C), clear(E)$

